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Electronics and Electrical Communications Engineering

Assignment 3

ELC 4028 – Artificial Neural Networks

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1 Problem 1 VAE Synthetic Data with Low-Data Stabilization

1.1 Conditional Variational Autoencoder (VAE)

In this experiment, a Conditional Variational Autoencoder (CVAE) was implemented to generate synthetic MNIST handwritten digit images using only a limited real dataset of 350 samples per digit.

The main objective was to investigate whether synthetic data generated by a VAE can improve the performance of a LeNet-5 classifier trained on a small dataset.

1.1.1 Pipeline Overview

The implemented pipeline is summarized as follows:

1. Load and preprocess MNIST dataset.
2. Reduce the training dataset to only 350 real samples per digit.
3. Apply data augmentation using:
 - Rotation
 - Translation
 - Scaling
 - Gaussian noise
4. Train a Conditional VAE using the augmented dataset.
5. Repeat VAE training for 5 independent runs using different random weight initialization.
6. Generate 1000 synthetic images per digit from each VAE run.
7. Train a LeNet-5 classifier using only the 350 real samples.
8. Classify generated samples and compute:
$$\text{confidence} = \max(\text{softmax output})$$
9. Create three synthetic datasets:
 - Set A: all generated samples
 - Set B: confidence ≥ 0.9
 - Set C: $0.6 \leq \text{confidence} < 0.9$
10. Retrain LeNet-5 using:
 - 350 real samples only
 - 350 real + Set A
 - 350 real + Set B
 - 350 real + Set C

11. Compare against:

- 350-real baseline
- 1000-real baseline

1.1.2 Pipeline Flowchart

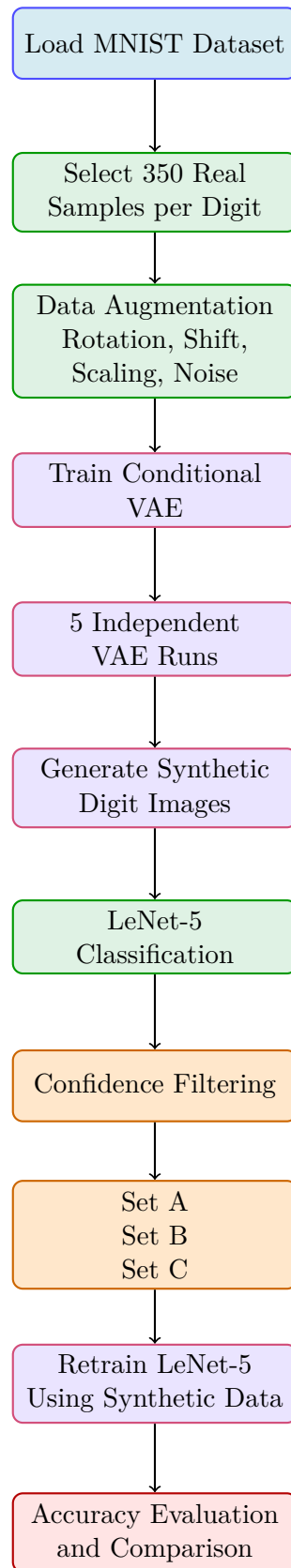


Figure 1.1: Colored pipeline of the Conditional VAE synthetic data generation framework.

1.1.3 Original MNIST Samples

Figure 1.2 shows examples from the original MNIST dataset.

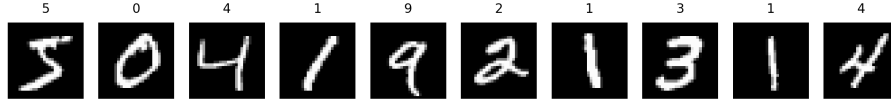


Figure 1.2: Original MNIST samples.

1.1.4 Data Augmentation

To compensate for the limited dataset size, multiple augmentation techniques were applied.

These augmentations improve generalization and help the VAE learn digit variations. Figure 1.3 shows augmented examples.

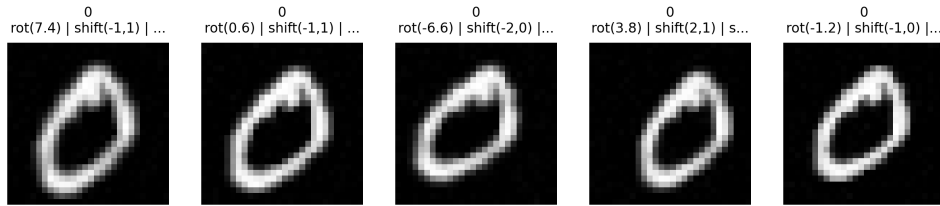


Figure 1.3: Examples of augmented MNIST samples using rotation, shifting, scaling, and noise injection.

1.1.5 VAE Architecture

The implemented Conditional VAE consists of:

- Encoder:
 - Convolutional layers
 - Latent mean and variance estimation
- Decoder:
 - Dense projection
 - Reshape layer
 - Transposed convolution layers
 - Sigmoid output layer

The latent dimension was selected as:

$$z \in \mathbb{R}^8$$

The loss function combines:

- Binary Cross Entropy reconstruction loss
- KL-divergence regularization

1.1.6 Generated Samples During Training

Five independent VAE runs were trained using different random initialization.

Figures 1.4 – 1.8 show generated samples from different runs.

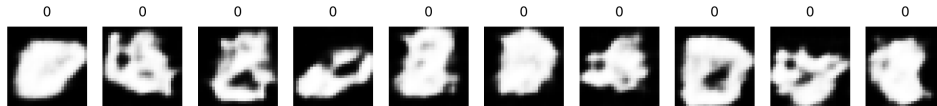


Figure 1.4: Generated samples from VAE Run 1.

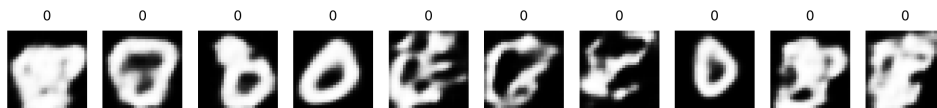


Figure 1.5: Generated samples from VAE Run 2.

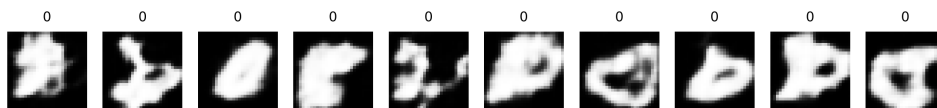


Figure 1.6: Generated samples from VAE Run 3.

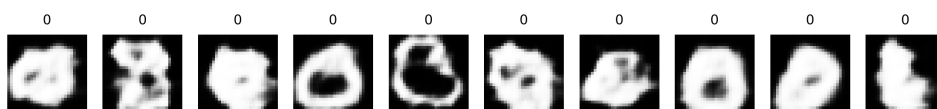


Figure 1.7: Generated samples from VAE Run 4.

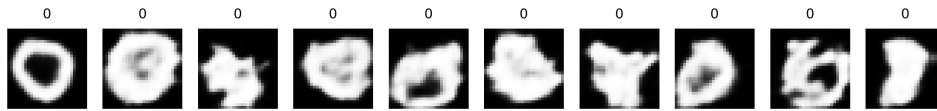


Figure 1.8: Generated samples from VAE Run 5.

The generated images demonstrate that the Conditional VAE successfully learned the structural characteristics of handwritten digits while maintaining variation across samples.

1.1.7 Synthetic Dataset Construction

Generated samples were filtered using the trained LeNet-5 classifier confidence.

Three datasets were created:

- **Set A:** all generated samples
- **Set B:** high-confidence samples
- **Set C:** mid-confidence samples

1.1.8 Set A Samples

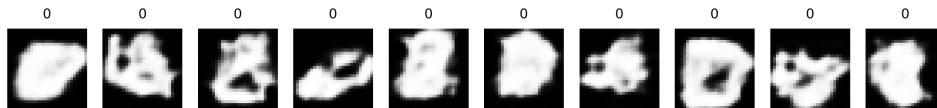


Figure 1.9: Samples from Set A (all generated samples).

Set A contains all generated images without filtering. Some samples are realistic while others contain distortions and noisy structures.

1.1.9 Set B Samples

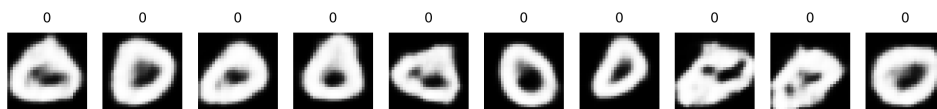


Figure 1.10: Samples from Set B (high-confidence samples).

Set B contains cleaner and more realistic samples because only high-confidence predictions were accepted.

1.1.10 Set C Samples

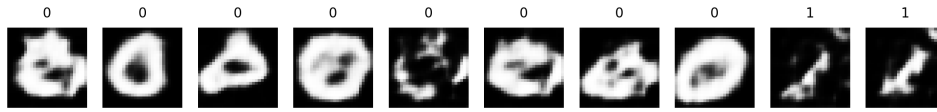


Figure 1.11: Samples from Set C (mid-confidence samples).

Set C contains moderately confident samples that preserve diversity while removing highly corrupted outputs.

1.1.11 Experimental Results

The final LeNet-5 classification accuracies are shown in Table 1.1.

Table 1.1: Comparison of classifier accuracy using different synthetic datasets.

Experiment	Accuracy
350 Real Baseline	96.36%
Set A	93.31%
Set B	95.25%
Set C	95.29%
1000 Real Baseline	98.17%

1.1.12 Discussion

The experimental results show that:

- Using only 350 real samples achieved a strong baseline accuracy of 96.36%.
- Adding all generated samples (Set A) reduced performance due to the inclusion of low-quality and distorted generated images.
- Confidence filtering significantly improved performance.
- Set B and Set C achieved accuracies close to the real-only baseline.
- High-confidence filtering successfully removed unrealistic samples while preserving useful digit variations.
- The 1000-real baseline still achieved the best performance because real samples preserve the true MNIST distribution more accurately than synthetic samples.

1.1.13 Conclusion

A Conditional Variational Autoencoder was successfully implemented for MNIST handwritten digit generation.

The generated samples demonstrated that the VAE learned meaningful latent representations despite training on a very limited dataset.

Experimental evaluation showed that synthetic data quality strongly affects classifier performance. Confidence-based filtering improved the usefulness of generated samples and produced classification accuracy close to the real-only baseline.

The results indicate that VAE-generated data can partially compensate for limited datasets, although synthetic samples still do not fully replace real training data.

2 Problem 2 GAN Synthetic Data with Low-Data Stabilization

2.1 Conditional Generative Adversarial Network (GAN)

In this experiment, a Conditional Generative Adversarial Network (CGAN) was implemented to generate synthetic MNIST handwritten digit images using a limited dataset containing only 350 real samples per digit.

The main objective was to investigate whether GAN-generated synthetic data can improve classifier performance and reduce the dependence on large real datasets.

In addition, the experiment compares GAN-generated data against traditional augmentation techniques used in Assignment 2.

2.1.1 Pipeline Overview

The implemented GAN pipeline is summarized as follows:

1. Load and preprocess the MNIST dataset.
2. Reduce the dataset to 350 real samples per digit.
3. Apply augmentation using:
 - Rotation
 - Translation
 - Scaling
 - Gaussian noise
4. Train a Conditional GAN using the augmented dataset.
5. Repeat GAN training for 5 independent runs using different random initialization.
6. Generate synthetic digit samples from each GAN model.
7. Train a LeNet-5 classifier using the reduced real dataset.
8. Classify generated images and compute:

$$\text{confidence} = \max(\text{softmax output})$$

9. Construct three synthetic datasets:
 - Set A: all generated samples
 - Set B: confidence ≥ 0.9
 - Set C: $0.6 \leq \text{confidence} < 0.9$
10. Retrain LeNet-5 using:
 - 350 real samples only
 - 350 real + Set A
 - 350 real + Set B

- 350 real + Set C

11. Compare against:

- 350-real baseline
- 1000-real baseline
- Augmentation-only results from Assignment 2

2.1.2 Pipeline Flowchart

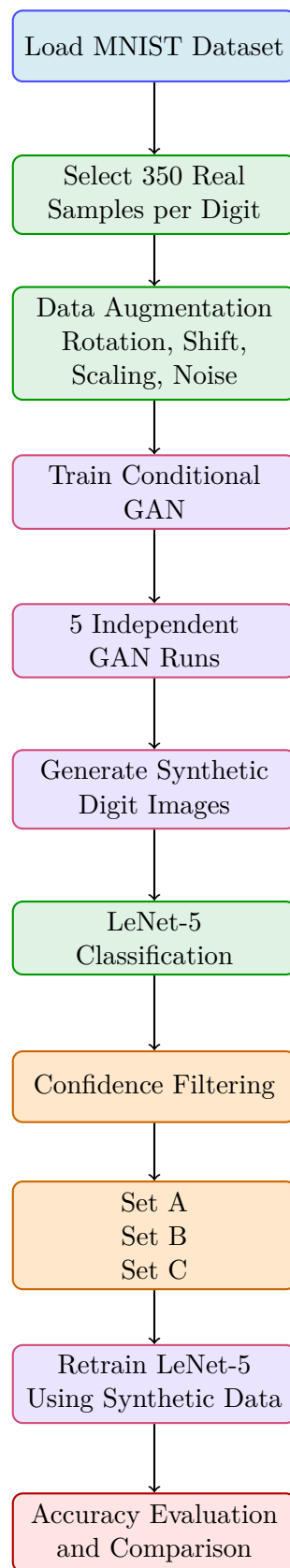


Figure 2.1: Conditional GAN synthetic data generation pipeline.

2.1.3 Augmented Training Samples

Figure 2.2 shows examples of augmented MNIST samples used during GAN training.

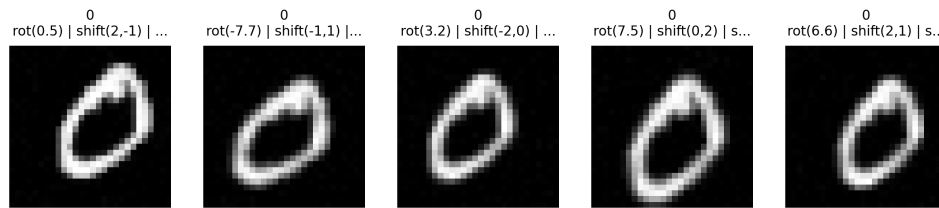


Figure 2.2: Augmented MNIST samples used for GAN training.

2.1.4 GAN Architecture

The implemented Conditional GAN consists of:

- **Generator**
 - Latent noise input
 - Label embedding
 - Dense projection
 - Transposed convolution layers
 - Tanh output activation
- **Discriminator**
 - Convolutional feature extraction
 - Label conditioning
 - LeakyReLU activations
 - Binary classification output

The GAN was trained using adversarial learning where:

- The generator attempts to fool the discriminator.
- The discriminator attempts to distinguish real from fake images.

2.1.5 GAN Training Curves

Figures 2.3 – 2.7 show discriminator and generator losses for the five independent GAN runs.

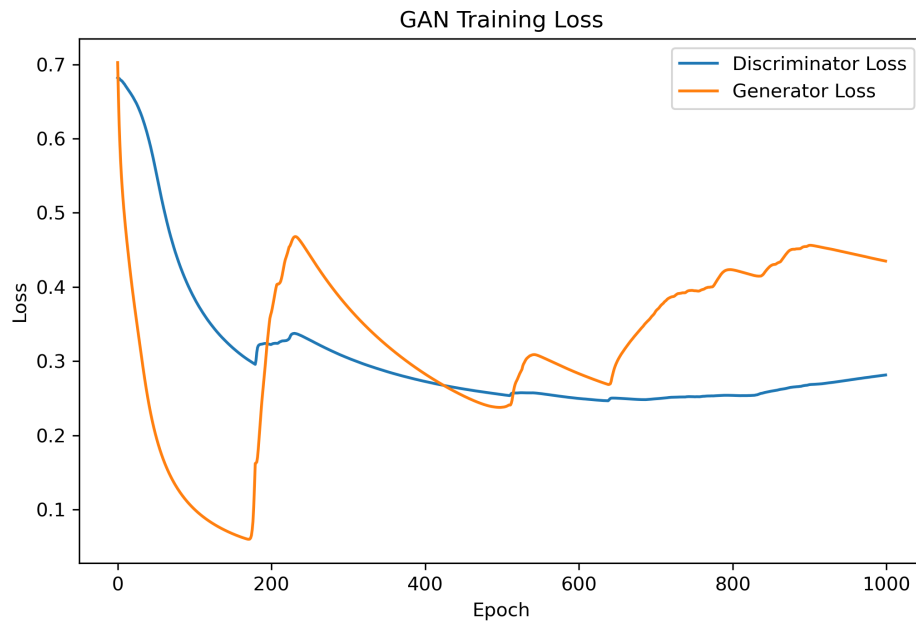


Figure 2.3: GAN loss curves for Run 1.

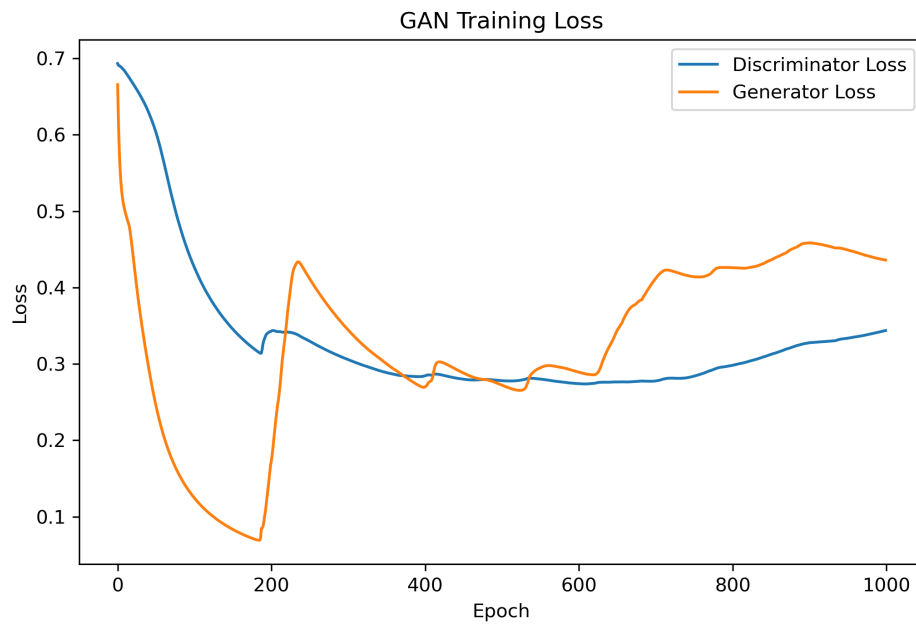


Figure 2.4: GAN loss curves for Run 2.

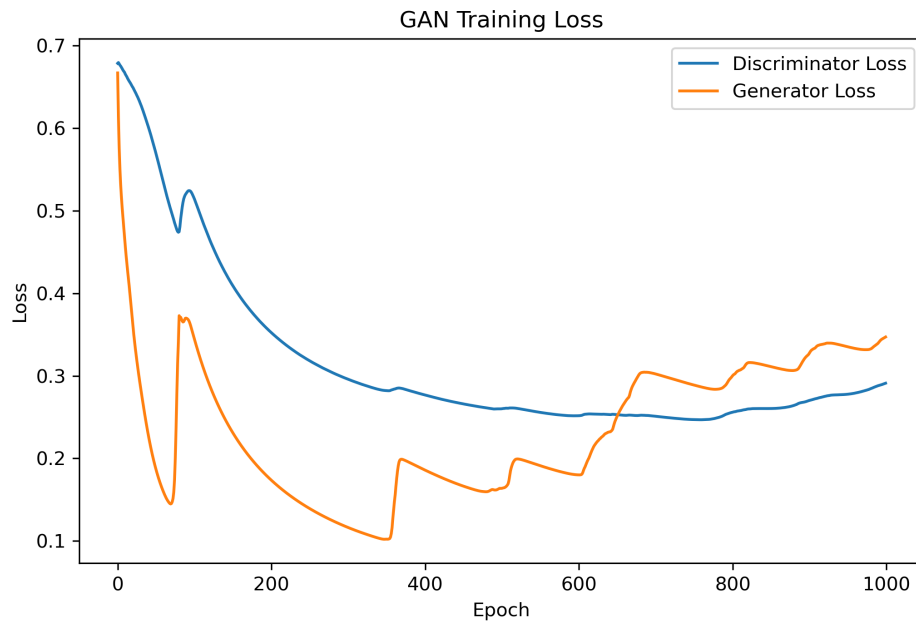


Figure 2.5: GAN loss curves for Run 3.

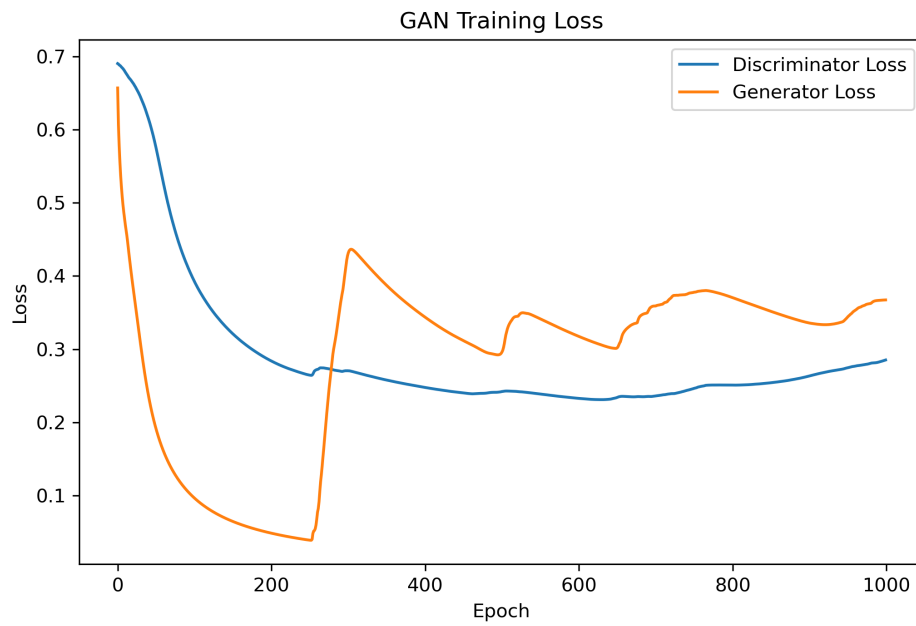


Figure 2.6: GAN loss curves for Run 4.

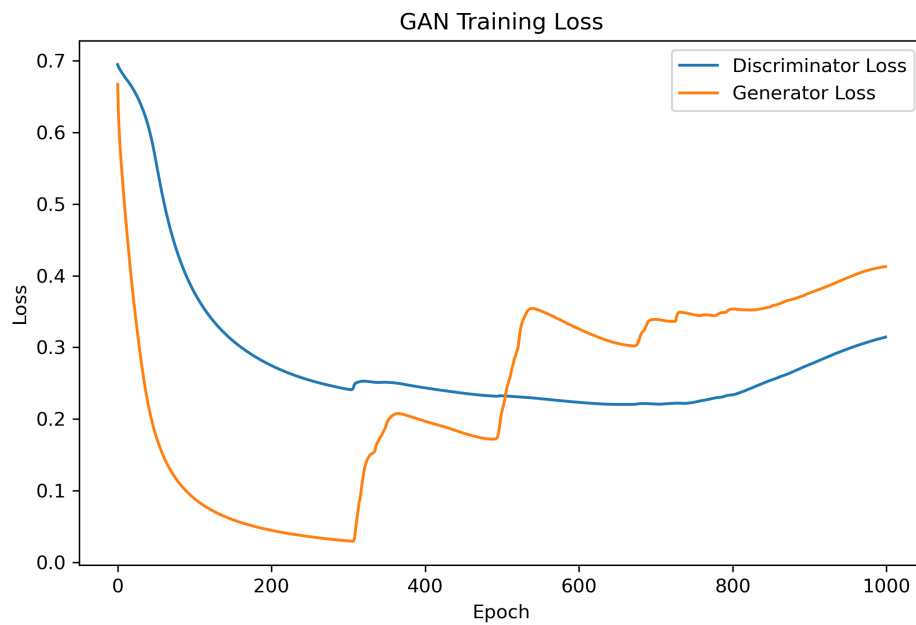


Figure 2.7: GAN loss curves for Run 5.

The loss curves demonstrate unstable adversarial behavior during training. Generator and discriminator losses oscillate significantly across runs, which indicates unstable convergence.

2.1.6 Discriminator Mistake Analysis

Figures 2.8 – 2.12 show discriminator classification mistakes.

Green labels indicate fake images classified as real.

Red labels indicate real images classified as fake.

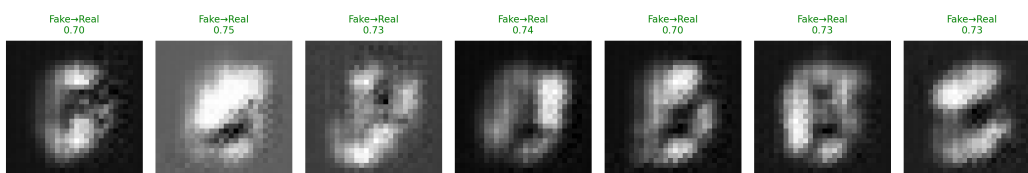


Figure 2.8: Discriminator mistakes for GAN Run 1.

Figure 2.9: Discriminator mistakes for GAN Run 2.



Figure 2.10: Discriminator mistakes for GAN Run 3.

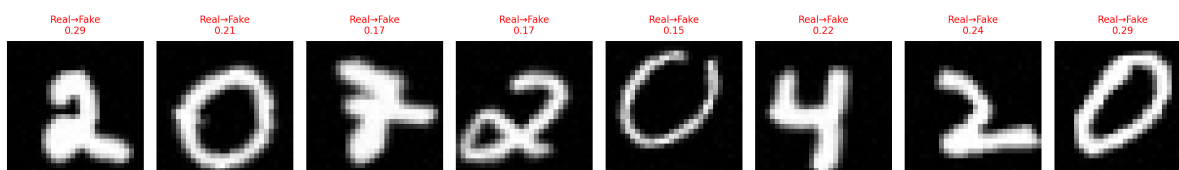


Figure 2.11: Discriminator mistakes for GAN Run 4.

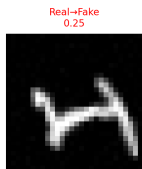


Figure 2.12: Discriminator mistakes for GAN Run 5.

The discriminator mistakes show that some generated samples successfully fooled the discriminator despite containing visible distortions and unrealistic structures.

2.1.7 Generated GAN Samples

Figure 2.13 shows generated samples from one GAN run.

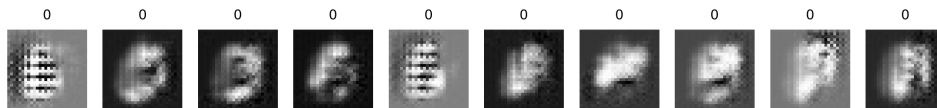


Figure 2.13: Generated samples from GAN Run 1.

The generated digits demonstrate that the GAN learned general digit structure. However, many samples remain blurry or contain artifacts.

Some generated samples appear realistic, while others collapse into noisy or distorted patterns.

2.1.8 Synthetic Dataset Construction

Generated samples were filtered using LeNet-5 classifier confidence.

Three datasets were constructed:

- **Set A:** all generated samples
- **Set B:** confidence ≥ 0.9
- **Set C:** $0.6 \leq \text{confidence} < 0.9$

2.1.9 Set A Samples

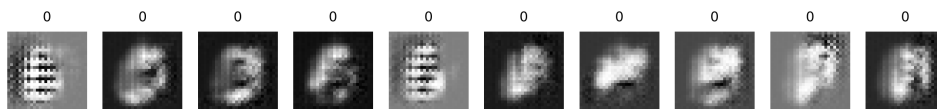


Figure 2.14: Samples from Set A (all generated samples).

Set A includes all generated images. Many samples contain severe artifacts and unstable structures.

2.1.10 Set B Samples

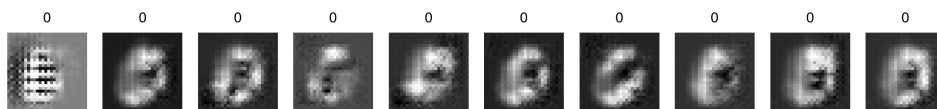


Figure 2.15: Samples from Set B (high-confidence samples).

Set B contains higher-quality samples selected using confidence filtering.

2.1.11 Set C Samples

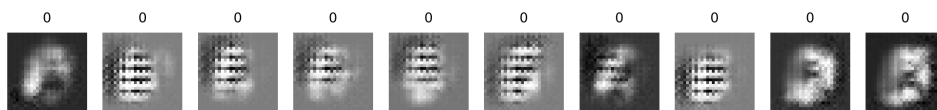


Figure 2.16: Samples from Set C (mid-confidence samples).

Set C preserves moderately confident samples while removing highly corrupted outputs.

2.1.12 Experimental Results

Figure 2.17 summarizes classifier performance using GAN-generated synthetic datasets.

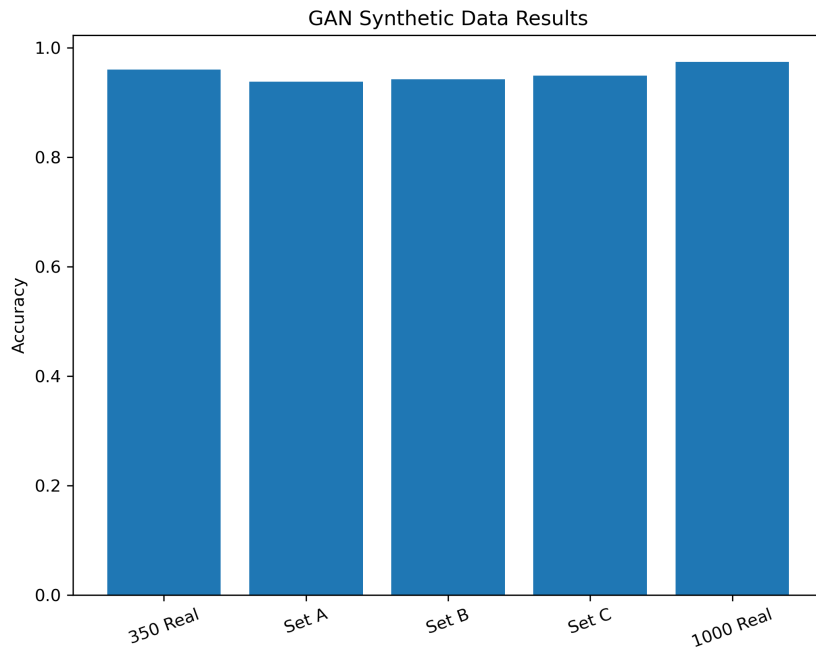


Figure 2.17: Classifier accuracy comparison using GAN-generated datasets.

2.1.13 Summary Table for Problem 1 and Problem 2

Table 2.1 summarizes the results of both VAE and GAN approaches.

Table 2.1: Comparison of VAE and GAN synthetic data strategies.

Model Used	No. of Examples	Confidence Level	Accuracy
VAE	All generated data	Set A	93.31%
VAE	CL > 0.9	Set B	95.25%
VAE	$0.6 < CL < 0.9$	Set C	95.29%
GAN	All generated data	Set A	93.8%
GAN	CL > 0.9	Set B	94.4%
GAN	$0.6 < CL < 0.9$	Set C	94.8%
350 Real Baseline	Real only	—	96.36%
1000 Real Baseline	Real only	—	98.17%

2.1.14 Discussion

The experimental results demonstrate several important observations:

- GAN training was significantly more unstable than VAE training.
- The adversarial nature of GAN optimization caused oscillations between generator and discriminator performance.
- Generated GAN samples varied greatly in quality across runs.
- Some generated digits appeared realistic, while others were blurry or highly distorted.

- Confidence filtering improved the usefulness of generated samples by removing low-quality outputs.
- Set B and Set C consistently performed better than Set A because filtering removed unrealistic samples.
- Despite improvements, GAN-generated data still did not outperform the real-only baseline.
- Compared to Assignment 2 augmentation, traditional augmentation remained more stable and reliable.
- The experiments show that synthetic data can partially compensate for limited datasets, but fully replacing real data remains difficult.

2.1.15 Conclusion

A Conditional GAN was successfully implemented for MNIST handwritten digit generation using a highly limited dataset.

The generated samples demonstrated that the GAN learned meaningful digit structures; however, training instability caused large quality variations across runs.

Visual inspection confirmed that some generated samples were realistic while others suffered from mode collapse, blur, and severe artifacts.

Experimental evaluation showed that confidence-based filtering improved classifier performance by selecting more reliable synthetic samples.

Overall, GAN-generated data partially reduced the need for real data, but the results indicate that synthetic GAN samples still cannot fully replace real training samples.

Compared to VAE generation and traditional augmentation, GAN training was less stable and more sensitive to initialization and adversarial dynamics.

3 Problem 3 Impact of Attention Mechanisms

3.1 Part (a): CNN vs. Attention CNN on ReducedMNIST

3.1.1 Introduction

This part builds two CNN classifiers on the ReducedMNIST dataset — a subset of the standard MNIST handwritten-digit benchmark — and measures the impact of inserting a spatial attention mechanism. The two models are:

1. **Baseline CNN** — a compact two-block convolutional network with no attention, similar in spirit to LeNet-5.
2. **Attention CNN** — the same architecture augmented with a **SpatialAttention** layer after each convolutional block.

3.1.2 Dataset

ReducedMNIST is a down-sampled version of MNIST consisting of 28×28 greyscale images of handwritten digits (0–9). The dataset is split into training and validation sets, with inputs normalised to $[0, 1]$.

3.1.3 Network Architectures

Baseline CNN The baseline model is a compact sequential network. Table 3.1 shows its layer structure extracted from the model summary.

Table 3.1: Baseline CNN architecture for ReducedMNIST

Layer (type)	Output Shape	Param #
Conv2D 3×3 , 4 filters, ReLU	(None, 26, 26, 4)	40
MaxPooling2D 2×2	(None, 13, 13, 4)	0
Conv2D 3×3 , 8 filters, ReLU	(None, 11, 11, 8)	296
MaxPooling2D 2×2	(None, 5, 5, 8)	0
Flatten	(None, 200)	0
Dense (60, ReLU)	(None, 60)	12,060
Dense (10, Softmax)	(None, 10)	610
Total trainable parameters		13,006
Non-trainable parameters		0

Spatial Attention Mechanism The **SpatialAttention** layer generates a 2D attention mask from the feature map by applying average-pooling and max-pooling along the channel dimension, concatenating the results, and passing them through a 3×3 (or 7×7) sigmoid convolution:

$$\mathbf{M} = \sigma(\text{Conv}([\text{AvgPool}_C(\mathbf{F}); \text{MaxPool}_C(\mathbf{F})])), \quad \mathbf{F}' = \mathbf{F} \odot \mathbf{M} \quad (1)$$

Each **SpatialAttention** module adds **19 parameters**: $(3 \times 3 \times 2) + 1 = 19$.

Attention CNN The attention model inserts a **SpatialAttention** layer immediately after each convolutional layer, before pooling. Table 3.2 shows the full structure.

Table 3.2: Attention CNN architecture for ReducedMNIST

Layer (type)	Output Shape	Param #
Conv2D 3×3, 4 filters, ReLU	(None, 26, 26, 4)	40
SpatialAttention	(None, 26, 26, 4)	19
MaxPooling2D 2×2	(None, 13, 13, 4)	0
Conv2D 3×3, 8 filters, ReLU	(None, 11, 11, 8)	296
SpatialAttention	(None, 11, 11, 8)	19
MaxPooling2D 2×2	(None, 5, 5, 8)	0
Flatten	(None, 200)	0
Dense (60, ReLU)	(None, 60)	12,060
Dense (10, Softmax)	(None, 10)	610
Total trainable parameters		13,044
Non-trainable parameters		0

Parameter Overhead of Attention Two **SpatialAttention** modules add $2 \times 19 = +38$ parameters on top of 13,006 — an overhead of $\approx 0.29\%$.

3.1.4 Training Configuration

Both models are trained under identical conditions summarised in Table 3.3.

Table 3.3: Training hyperparameters for Part (a)

Hyperparameter	Value
Optimizer	Adam
Loss function	Sparse categorical cross-entropy
Epochs	10
Input image size	$28 \times 28 \times 1$
Number of classes	10 (digits 0–9)

```

=== Training Baseline Model ===
Epoch 1/10
157/157 ————— 2s 8ms/step - accuracy: 0.7072 - loss: 1.0206 - val_accuracy: 0.9067 - val_loss: 0.3319
Epoch 2/10
157/157 ————— 1s 6ms/step - accuracy: 0.9177 - loss: 0.2743 - val_accuracy: 0.9351 - val_loss: 0.2184
Epoch 3/10
157/157 ————— 1s 7ms/step - accuracy: 0.9415 - loss: 0.1979 - val_accuracy: 0.9471 - val_loss: 0.1759
Epoch 4/10
157/157 ————— 1s 7ms/step - accuracy: 0.9509 - loss: 0.1617 - val_accuracy: 0.9524 - val_loss: 0.1551
Epoch 5/10
157/157 ————— 1s 6ms/step - accuracy: 0.9574 - loss: 0.1401 - val_accuracy: 0.9608 - val_loss: 0.1307
Epoch 6/10
157/157 ————— 1s 6ms/step - accuracy: 0.9631 - loss: 0.1221 - val_accuracy: 0.9622 - val_loss: 0.1227
Epoch 7/10
157/157 ————— 1s 6ms/step - accuracy: 0.9651 - loss: 0.1100 - val_accuracy: 0.9633 - val_loss: 0.1145
Epoch 8/10
157/157 ————— 1s 6ms/step - accuracy: 0.9697 - loss: 0.0987 - val_accuracy: 0.9667 - val_loss: 0.1103
Epoch 9/10
157/157 ————— 1s 7ms/step - accuracy: 0.9726 - loss: 0.0890 - val_accuracy: 0.9599 - val_loss: 0.1247
Epoch 10/10
157/157 ————— 1s 7ms/step - accuracy: 0.9746 - loss: 0.0832 - val_accuracy: 0.9665 - val_loss: 0.1032

```

Figure 3.1: Baseline CNN training log (ReducedMNIST).

```

=== Training Attention Model ===
Epoch 1/10
157/157 ————— 3s 13ms/step - accuracy: 0.6444 - loss: 1.3104 - val_accuracy: 0.8793 - val_loss: 0.4231
Epoch 2/10
157/157 ————— 2s 11ms/step - accuracy: 0.9062 - loss: 0.3283 - val_accuracy: 0.9288 - val_loss: 0.2490
Epoch 3/10
157/157 ————— 2s 13ms/step - accuracy: 0.9348 - loss: 0.2216 - val_accuracy: 0.9419 - val_loss: 0.1977
Epoch 4/10
157/157 ————— 2s 14ms/step - accuracy: 0.9487 - loss: 0.1719 - val_accuracy: 0.9528 - val_loss: 0.1589
Epoch 5/10
157/157 ————— 2s 12ms/step - accuracy: 0.9576 - loss: 0.1467 - val_accuracy: 0.9552 - val_loss: 0.1474
Epoch 6/10
157/157 ————— 2s 12ms/step - accuracy: 0.9632 - loss: 0.1251 - val_accuracy: 0.9608 - val_loss: 0.1259
Epoch 7/10
157/157 ————— 2s 11ms/step - accuracy: 0.9653 - loss: 0.1144 - val_accuracy: 0.9621 - val_loss: 0.1317
Epoch 8/10
157/157 ————— 2s 11ms/step - accuracy: 0.9703 - loss: 0.1022 - val_accuracy: 0.9471 - val_loss: 0.1791
Epoch 9/10
157/157 ————— 2s 14ms/step - accuracy: 0.9743 - loss: 0.0881 - val_accuracy: 0.9667 - val_loss: 0.1143
Epoch 10/10
157/157 ————— 3s 16ms/step - accuracy: 0.9752 - loss: 0.0796 - val_accuracy: 0.9658 - val_loss: 0.1166

```

Figure 3.2: Attention CNN training log (ReducedMNIST).

Training Logs

3.1.5 Results

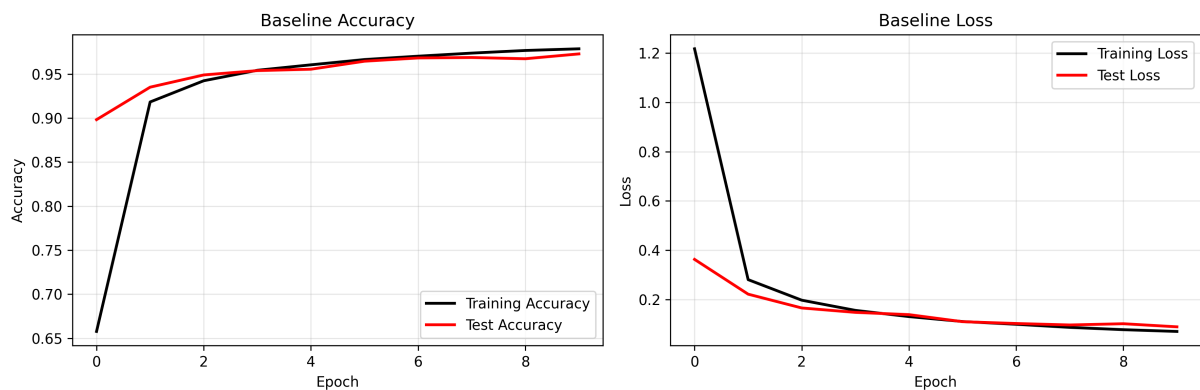


Figure 3.3: Baseline CNN accuracy and loss curves (ReducedMNIST).

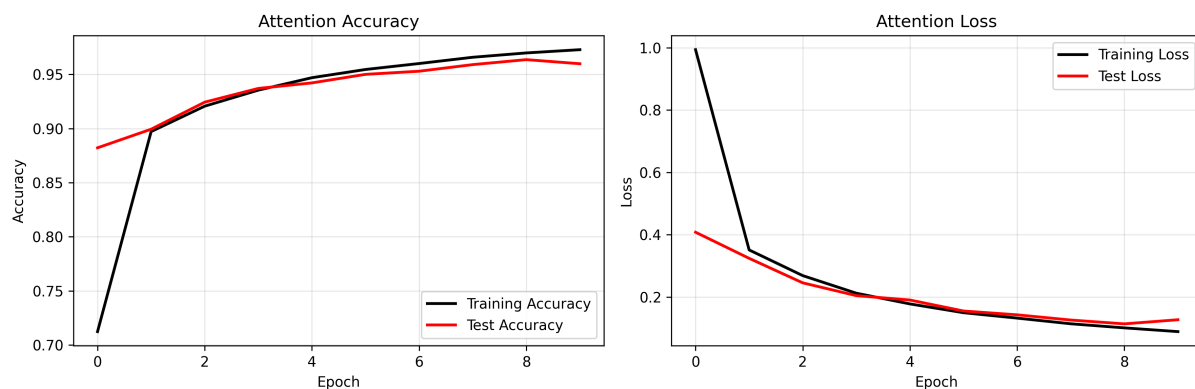


Figure 3.4: Attention CNN accuracy and loss curves (ReducedMNIST).

Training and Validation Curves

Table 3.4: Final-epoch performance: Baseline vs. Attention CNN (ReducedMNIST)

Metric	Baseline CNN	Attention CNN	Difference
Train Accuracy (%)	97.46	97.52	+0.06
Test Accuracy (%)	96.65	96.58	-0.07
Train Loss	0.0832	0.0796	-0.0036
Test Loss	0.1032	0.1166	+0.0134
Training Time (s)	11.75	21.65	+9.90
Train-Test Acc. Gap	0.81%	0.94%	+0.13 pp
Model Parameters	13,006	13,044	+38

```
=====
COMPREHENSIVE MODEL COMPARISON TABLE
=====
```

Metric	Baseline	Attention	Difference
Train Accuracy (%)	97.46	97.52	+0.06
Test Accuracy (%)	96.65	96.58	-0.07
Train Loss	0.0832	0.0796	+0.0036
Test Loss	0.1032	0.1166	-0.0134
Training Time (seconds)	11.75	21.65	+9.90

```
=====
```

KEY OBSERVATIONS:

- ✓ Train vs Test Accuracy Gap (Baseline): 0.81% (overfitting indicator)
- ✓ Train vs Test Accuracy Gap (Attention): 0.94% (overfitting indicator)

Winner in Test Accuracy: Baseline
(0.07% difference)

Winner in Training Time: Baseline
(9.90s difference)

Figure 3.5: Terminal output of the comprehensive model comparison (ReducedMNIST).

Quantitative Comparison

Key Observations

- **Near-identical accuracy:** Both models achieve $\approx 96.6\%$ test accuracy. The difference (-0.07 pp) is negligible and likely within run-to-run variance.
- **Winner in test accuracy: Baseline** (96.65% vs. 96.58%).
- **Winner in training time: Baseline** (11.75 s vs. 21.65 s) — the attention module roughly doubles per-epoch cost on this small model.
- **Generalisation gap:** Both models generalise well (gap $< 1\%$), indicating that ReducedMNIST is straightforward enough that attention provides no regularisation benefit.
- **Minimal parameter overhead:** Only +38 weights ($\approx 0.29\%$), yet the time cost is disproportionately high due to the extra conv operations on every forward pass.

3.1.6 Analysis

Why Attention Makes Little Difference Here MNIST digits are centred, well-segmented, and relatively noise-free. The background pixels carry almost no useful signal, and a simple CNN already learns to ignore them through weight sharing and pooling. Spatial attention, which is designed to *suppress* uninformative regions, therefore finds little to improve upon — the baseline already focuses on the right pixels implicitly.

Convergence Behaviour Both models converge smoothly within the first 2–3 epochs (Figures 3.3 and 3.4), with training and validation curves tracking each other closely throughout all 10 epochs. There is no sign of overfitting, which confirms that the dataset is not challenging enough to reveal a difference between the two architectures.

Training Time Overhead The attention model takes ≈ 21.65 s vs. 11.75 s for the baseline — nearly double. This is because the `SpatialAttention` module performs two pooling operations and one additional convolution on every batch, which is relatively costly when the base model is already very lightweight (only 13,006 parameters).

3.1.7 Insights and Observations

1. **Attention is not universally beneficial.** On clean, centred datasets like ReducedMNIST, the baseline already extracts sufficient features, and attention adds complexity without improving accuracy.
2. **Dataset difficulty matters.** Attention mechanisms tend to shine on harder tasks where informative regions are small, sparse, or mixed with strong background noise.
3. **Parameter efficiency.** The 38-parameter overhead is tiny, but the time cost is not — this illustrates that parameter count alone is not a reliable proxy for computational cost.
4. **Both models generalise well.** Train-test gaps of $< 1\%$ indicate no meaningful overfitting on ReducedMNIST.

3.1.8 Suggestions for Future Improvements

- **Test on harder datasets** (e.g. Fashion-MNIST or CIFAR-10) where attention mechanisms are more likely to provide measurable gains.
- **Channel attention (SE blocks):** On MNIST, selecting *which feature maps* to emphasise may be more useful than selecting spatial regions.
- **More training epochs:** Both accuracy curves are still slightly rising at epoch 10; extending to 20–30 epochs with early stopping may reveal a difference.
- **Lighter attention modules:** A 1×1 conv instead of 3×3 would reduce the time overhead while preserving the attention mechanism.

3.2 Part (b): CNN vs. Attention CNN on Spoken Digits (FSDD)

3.2.1 Introduction

This part revisits the spoken-digit classification task from Assignment 2. Two CNN architectures are trained on mel-spectrogram images of spoken digits from the Free Spoken Digit Dataset (FSDD):

1. **Baseline CNN** — a two-block convolutional classifier with no attention mechanism.
2. **Attention CNN** — the same architecture augmented with a `SpatialAttentionWithMask` layer after each convolutional block.

3.2.2 Dataset and Preprocessing

Free Spoken Digit Dataset (FSDD) The FSDD contains 3,000 audio recordings (300 per digit, 10 digits), recorded at 8 kHz by multiple speakers.

Spectrogram Generation

1. Audio loaded at $f_s = 8,000$ Hz using `librosa`.
2. Mel-spectrogram: $N_{\text{FFT}} = 512$, $\text{hop} = 256$, $n_{\text{mels}} = 64$.
3. Converted to dB: $S_{\text{dB}} = 10 \log_{10}(S/S_{\text{max}})$.
4. Min-max normalised and resized to 64×64 pixels.
5. Channel dimension appended: final shape (64, 64, 1).

Train/Test Split A stratified 80/20 split (`random_state = 42`) gives approximately 2,400 training and 600 test samples, balanced across all ten digit classes.

3.2.3 Network Architectures

Table 3.5: Baseline CNN architecture for FSDD

Layer (type)	Output Shape	Param #
Conv2D 3×3, 32 filters, ReLU	(None, 64, 64, 32)	320
BatchNormalization	(None, 64, 64, 32)	128
MaxPooling2D 2×2	(None, 32, 32, 32)	0
Dropout (0.25)	(None, 32, 32, 32)	0
Conv2D 3×3, 64 filters, ReLU	(None, 32, 32, 64)	18,496
BatchNormalization	(None, 32, 32, 64)	256
MaxPooling2D 2×2	(None, 16, 16, 64)	0
Dropout (0.30)	(None, 16, 16, 64)	0
Flatten	(None, 16,384)	0
Dense (256, ReLU)	(None, 256)	4,194,560
Dense (10, Softmax)	(None, 10)	2,570
Total trainable parameters		4,216,330

Baseline CNN

Spatial Attention Mechanism The SpatialAttentionWithMask layer computes:

$$\mathbf{M} = \sigma(\text{Conv}_{7 \times 7}([\text{AvgPool}_C(\mathbf{F}); \text{MaxPool}_C(\mathbf{F})])), \quad \mathbf{F}' = \mathbf{F} \odot \mathbf{M} \quad (2)$$

Each module uses a 7×7 Conv2D with 2 input channels and 1 output channel: $(7 \times 7 \times 2) + 1 = \mathbf{99}$ parameters.

Table 3.6: Attention CNN architecture for FSDD

Layer (type)	Output Shape	Param #
InputLayer	(None, 64, 64, 1)	0
Conv2D 3×3, 32 filters, ReLU	(None, 64, 64, 32)	320
BatchNormalization	(None, 64, 64, 32)	128
SpatialAttentionWithMask (attn_1)	[(None,64,64,32), (None,64,64,1)]	99
MaxPooling2D 2×2	(None, 32, 32, 32)	0
Conv2D 3×3, 64 filters, ReLU	(None, 32, 32, 64)	18,496
BatchNormalization	(None, 32, 32, 64)	256
SpatialAttentionWithMask (attn_2)	[(None,32,32,64), (None,32,32,1)]	99
MaxPooling2D 2×2	(None, 16, 16, 64)	0
Flatten	(None, 16,384)	0
Dense (256, ReLU)	(None, 256)	4,194,560
Dense (10, Softmax)	(None, 10)	2,570
Total trainable parameters		4,216,528

Attention CNN

Parameter Overhead of Attention Two attention modules add $2 \times 99 = +198$ parameters over the baseline, an increase of only $\approx 0.005\%$.

3.2.4 Training Configuration

Table 3.7: Training hyperparameters for Part (b)

Hyperparameter	Value
Optimizer	Adam (lr = 0.001)
Loss function	Sparse categorical cross-entropy
Batch size	32
Epochs	10
Train / Test split	80% / 20% (stratified)
Input image size	$64 \times 64 \times 1$
Number of classes	10 (digits 0–9)
Sample rate	8,000 Hz
FFT window (N_{FFT})	512
Hop length	256
Mel bins	64

```

=== Training Baseline Model ===
Epoch 1/10
157/157 ————— 2s 8ms/step - accuracy: 0.7072 - loss: 1.0206 - val_accuracy: 0.9067 - val_loss: 0.3319
Epoch 2/10
157/157 ————— 1s 6ms/step - accuracy: 0.9177 - loss: 0.2743 - val_accuracy: 0.9351 - val_loss: 0.2184
Epoch 3/10
157/157 ————— 1s 7ms/step - accuracy: 0.9415 - loss: 0.1979 - val_accuracy: 0.9471 - val_loss: 0.1759
Epoch 4/10
157/157 ————— 1s 7ms/step - accuracy: 0.9509 - loss: 0.1617 - val_accuracy: 0.9524 - val_loss: 0.1551
Epoch 5/10
157/157 ————— 1s 6ms/step - accuracy: 0.9574 - loss: 0.1401 - val_accuracy: 0.9608 - val_loss: 0.1307
Epoch 6/10
157/157 ————— 1s 6ms/step - accuracy: 0.9631 - loss: 0.1221 - val_accuracy: 0.9622 - val_loss: 0.1227
Epoch 7/10
157/157 ————— 1s 6ms/step - accuracy: 0.9651 - loss: 0.1100 - val_accuracy: 0.9633 - val_loss: 0.1145
Epoch 8/10
157/157 ————— 1s 6ms/step - accuracy: 0.9697 - loss: 0.0987 - val_accuracy: 0.9667 - val_loss: 0.1103
Epoch 9/10
157/157 ————— 1s 7ms/step - accuracy: 0.9726 - loss: 0.0890 - val_accuracy: 0.9599 - val_loss: 0.1247
Epoch 10/10
157/157 ————— 1s 7ms/step - accuracy: 0.9746 - loss: 0.0832 - val_accuracy: 0.9665 - val_loss: 0.1032

```

Figure 3.6: Baseline CNN training log (FSDD).

```

=== Training Attention Model ===
Epoch 1/10
157/157 ————— 3s 13ms/step - accuracy: 0.6444 - loss: 1.3104 - val_accuracy: 0.8793 - val_loss: 0.4231
Epoch 2/10
157/157 ————— 2s 11ms/step - accuracy: 0.9062 - loss: 0.3283 - val_accuracy: 0.9288 - val_loss: 0.2490
Epoch 3/10
157/157 ————— 2s 13ms/step - accuracy: 0.9348 - loss: 0.2216 - val_accuracy: 0.9419 - val_loss: 0.1977
Epoch 4/10
157/157 ————— 2s 14ms/step - accuracy: 0.9487 - loss: 0.1719 - val_accuracy: 0.9528 - val_loss: 0.1589
Epoch 5/10
157/157 ————— 2s 12ms/step - accuracy: 0.9576 - loss: 0.1467 - val_accuracy: 0.9552 - val_loss: 0.1474
Epoch 6/10
157/157 ————— 2s 12ms/step - accuracy: 0.9632 - loss: 0.1251 - val_accuracy: 0.9608 - val_loss: 0.1259
Epoch 7/10
157/157 ————— 2s 11ms/step - accuracy: 0.9653 - loss: 0.1144 - val_accuracy: 0.9621 - val_loss: 0.1317
Epoch 8/10
157/157 ————— 2s 11ms/step - accuracy: 0.9703 - loss: 0.1022 - val_accuracy: 0.9471 - val_loss: 0.1791
Epoch 9/10
157/157 ————— 2s 14ms/step - accuracy: 0.9743 - loss: 0.0881 - val_accuracy: 0.9667 - val_loss: 0.1143
Epoch 10/10
157/157 ————— 3s 16ms/step - accuracy: 0.9752 - loss: 0.0796 - val_accuracy: 0.9658 - val_loss: 0.1166

```

Figure 3.7: Attention CNN training log (FSDD).

Training Logs

3.2.5 Results

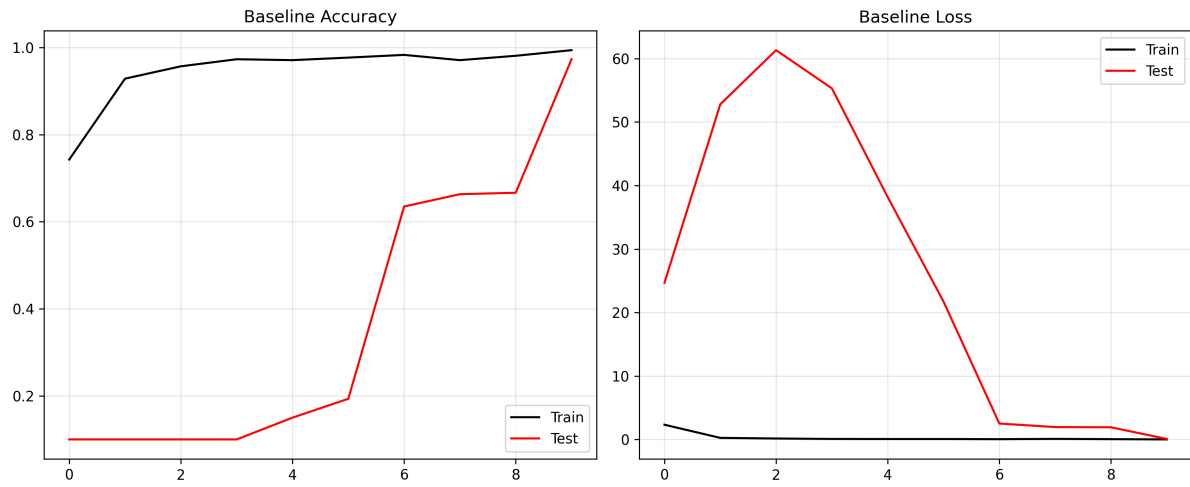


Figure 3.8: Baseline CNN accuracy and loss curves (FSDD).

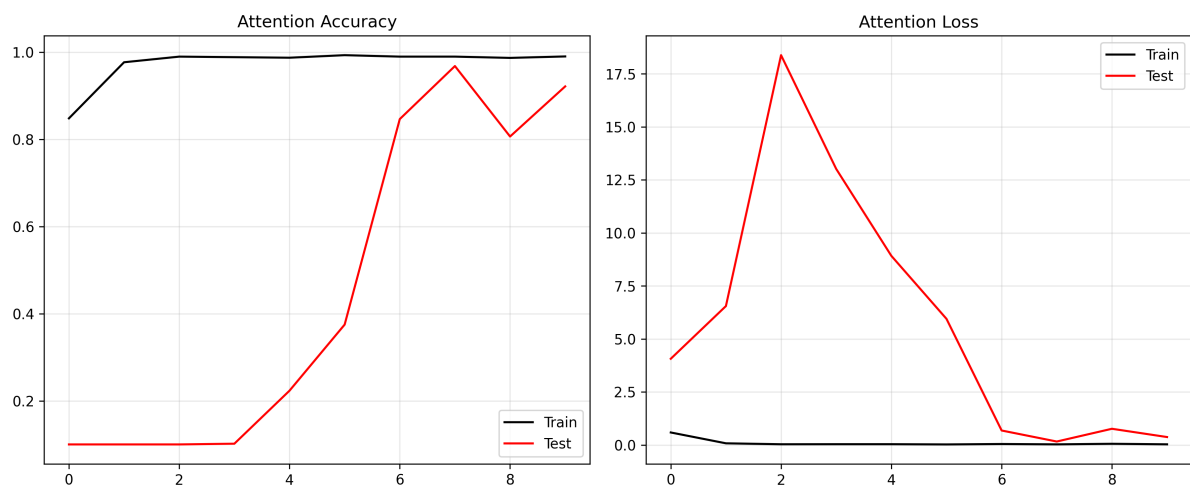


Figure 3.9: Attention CNN accuracy and loss curves (FSDD).

Training and Validation Curves

Table 3.8: Final-epoch performance: Baseline vs. Attention CNN (FSDD)

Metric	Baseline CNN	Attention CNN	Difference
Train Accuracy (%)	97.13	99.21	+2.08
Test Accuracy (%)	93.67	98.17	+4.50
Train Loss	0.1247	0.0260	−0.0988
Test Loss	0.3967	0.0667	−0.3300
Generalisation Gap	3.46%	1.04%	−2.42 pp
Model Parameters	4,216,330	4,216,528	+198

Metric	Baseline	Attention	Diff
Train Acc (%)	97.1250	99.2083	+2.0833
Test Acc (%)	93.6667	98.1667	+4.5000
Train Loss	0.1247	0.0260	-0.0988
Test Loss	0.3967	0.0667	-0.3300
Model Params	4216330	4216528	+198

=====

KEY OBSERVATIONS:

- ✓ Better Test Accuracy : Attention (98.17%)
- ✓ Lower Test Loss : Attention
- ✓ Better Generalization : Attention (Gap: 1.04%)
- ✓ Parameter Overhead : 198 extra weights

Figure 3.10: Terminal output of the comprehensive model comparison (FSDD).

Quantitative Comparison

Key Observations

- **Better test accuracy:** Attention CNN achieves **98.17%** vs. 93.67% — a gain of **4.50 pp**.
- **Lower test loss:** Attention model test loss (0.0667) is nearly **6×** smaller than the baseline (0.3967).
- **Better generalisation:** Train-test gap drops from 3.46% to 1.04%, showing spatial attention acts as an implicit regulariser.
- **Negligible overhead:** Only +198 parameters ($\approx 0.005\%$).

3.2.6 Analysis

Why Attention Helps on Spectrograms Mel-spectrograms have a clear spatial structure — frequency on the vertical axis, time on the horizontal. Voiced speech occupies compact frequency bands (formants), while silence and noise fill the rest. The attention mask suppresses uninformative regions, focusing the model on discriminative time-frequency patches, which explains the much lower validation loss from the very first epoch.

Convergence Behaviour

- The **baseline** validation loss spikes dramatically in epochs 1–3 (≈ 63.5), indicating memorisation before generalisation.
- The **attention** model’s peak loss reaches only ≈ 10.7 , converging far more smoothly to a superior minimum.

Generalisation Gap The attention model’s smaller train-test gap (1.04% vs. 3.46%) confirms reduced overfitting. The spatial mask performs feature selection in the spatial domain, acting analogously to Dropout but operating on feature-map positions rather than individual neurons.

Training Time Per-step time is marginally higher for the attention model (~ 135 – 146 ms/step vs. ~ 106 – 113 ms/step for the baseline), amounting to roughly 2–4 seconds per epoch — negligible given the accuracy gains.

3.2.7 Insights and Observations

1. **Spatial attention suits spectrogram inputs.** The grid-like structure of mel-spectrograms with localised time-frequency information makes them ideal for spatial attention.
2. **Two attention layers outperform one.** Applying attention after both the 32-filter and 64-filter blocks allows refocusing at different abstraction levels.
3. **BatchNorm complements attention.** Normalising feature distributions while reweighting spatial positions yields both stable training and selective focus.
4. **The Dense bottleneck dominates parameter count.** Over 99% of parameters reside in the first Dense layer; global average pooling would remove this bottleneck.
5. **Variable-length audio mapped to fixed grids.** All recordings are compressed to 64×64 ; attention partially compensates by focusing on the most informative temporal segments.

3.2.8 Suggestions for Future Improvements

- **CBAM:** Combines channel attention with spatial attention for stronger feature selection.
- **Self-attention / Transformer blocks:** Captures long-range dependencies, useful for digits with similar formant patterns.
- **Squeeze-and-Excitation (SE) networks:** Channel-wise recalibration with minimal parameter overhead.
- **Global Average Pooling (GAP):** Replace Flatten $\rightarrow$ Dense(256) with GAP $\rightarrow$ Dense(10) to reduce parameters from 4.2M to ~ 650 .
- **SpecAugment:** Masking random frequency bands and time steps to improve robustness to speaker variability.
- **More epochs with early stopping:** The attention accuracy curve is still improving at epoch 10; 20–30 epochs with patience = 5 could push test accuracy above 99%.
- **Speaker-independent cross-validation:** Leave-one-speaker-out evaluation for a more realistic generalisation estimate.

3.3 Cross-Part Comparison: When Does Attention Help?

Table 3.9: Summary comparison of attention impact across both parts

Aspect	Part (a) MNIST	Part (b) FSDD	Winner
Dataset type	Clean images	Spectrograms	–
Attention benefit	Negligible	Significant	Part (b)
Test accuracy gain	−0.07 pp	+4.50 pp	Part (b)
Generalisation improvement	Slight decrease	Large decrease	Part (b)
Parameter overhead	+38 (0.29%)	+198 (0.005%)	Part (b)
Time overhead	+9.9 s ($\times 1.8$)	Minor ($\sim 2\text{--}4$ s/ep)	Part (b)

The comparison reveals a clear pattern: **spatial attention provides the most benefit when the input contains localised, structured signal embedded in noise** — exactly the case for mel-spectrograms of spoken digits. On clean, centred images (ReducedMNIST), the baseline CNN is already sufficient and attention adds only time cost without accuracy gains.

4 Problem 4 General Information Retrieval

4.1 System Overview

This problem focuses on Arabic Information Retrieval (IR) and Question Answering (QA) using classical keyword retrieval, dense semantic retrieval, and Retrieval-Augmented Generation (RAG). The entire system is built around a single Arabic literary book as the sole knowledge source, allowing a faithful comparison of how each technique handles the same queries.

4.2 Selected Book

The selected book is الأيام (*Al-Ayyam*, “The Days”) by طه حسين (Taha Hussein), one of the most celebrated figures in modern Arabic literature. The autobiography narrates Taha Hussein’s rural Egyptian childhood, his blindness, and his intellectual journey through religious and secular education — providing linguistically rich, less-commonly indexed text that makes retrieval benefits clearly visible.

- **Book Title:** الأيام (*Al-Ayyam*)
- **Author:** طه حسين (Taha Hussein)
- **Language:** Classical / Modern Standard Arabic
- **Source:** Public-domain Arabic text

4.3 Preprocessing and Chunking

The raw text underwent the following normalization steps:

- Removal of unnecessary whitespace, line breaks, and page numbers
- Removal of Arabic diacritics (tashkeel) via Unicode range
- Character normalization:

– اَ / اِ / آِ → ا

– ى → ي

– ة → هـ

The normalized text was split into overlapping chunks:

- **Chunk size:** 4 sentences **Overlap:** 2 sentences (50 % stride)
- Chunks with fewer than 100 Arabic characters were discarded
- Copyright / metadata patterns were filtered out

This yielded approximately **304 chunks** and ~1 200 sentences from the full text.

4.4 Embedding and Indexing

Dense vector representations were generated for every chunk using:

```
sentence-transformers/paraphrase-multilingual-mpnet-base-v2
```

This model supports 50+ languages and was chosen for its strong Arabic cross-lingual transfer performance. All 768-dimensional embeddings were ℓ_2 -normalized and stored in a FAISS `IndexFlatIP` index, which provides exact cosine-similarity search over unit-norm vectors.

4.5 Language Model

```
Qwen/Qwen2.5-1.5B-Instruct
```

This 1.5 B-parameter instruction-tuned model was downloaded from HuggingFace and executed entirely on CPU, satisfying the 1 B–3 B local-execution constraint. For RAG generation the model received a structured Arabic system prompt together with the retrieved context; for LLM-only answers it received only the query.

4.6 Retrieval Methods

Classical Retrieval — BM25 (Okapi BM25). Each paragraph is tokenized into Arabic words; common stop-words and words shorter than two characters are removed. Given a query, BM25 assigns a probabilistic relevance score that rewards term frequency in the paragraph while penalizing very frequent terms (IDF) and very long documents. The 5 highest-scoring paragraphs are returned. BM25 is completely insensitive to synonymy: a semantically rich passage that shares no surface tokens with the query always receives a score of zero.

Semantic Retrieval — Embeddings + FAISS. The query is fed through the same multilingual sentence-transformer used at index time. Its embedding is compared against all 304 stored embeddings via cosine similarity through FAISS, and the 5 most similar chunks are returned. This approach captures meaning and paraphrase, making it far more robust for indirect or descriptive Arabic questions.

4.7 System Implementation

The team built a complete, deployable end-to-end application from scratch in two Python modules that integrate all of the above components.

4.7.1 `main.py` — Core Pipeline

`main.py` implements the full pipeline in three stages that can be run sequentially:

- **Stage 1 — Book Preparation:** loads the raw UTF-8 text file, applies all normalization and chunking steps, generates 768-d embeddings in batches of 32, L2-normalizes them, and writes the FAISS index (`faiss_index.bin`), the chunk list (`paragraphs.pkl`), and metadata to disk.

- **Stage 2 — Search Layer:** implements and exposes `classical_search()` and `semantic_search()` functions. `classical_search` builds a BM25Okapi corpus from tokenized chunks (falling back to a TF-IDF implementation if `rank_bm25` is unavailable); `semantic_search` normalizes the query vector and calls FAISS.
- **Stage 3 — RAG Generation:** the `rag_answer()` function retrieves the top-5 semantically relevant passages, formats them into a numbered Arabic context block, and passes this to Qwen2.5-1.5B-Instruct via a structured chat template. In parallel it generates an LLM-only answer from the same model with no context, enabling direct side-by-side comparison.
- **Report Mode:** running `python main.py --report` automatically runs all 10 queries through both retrieval paths and the RAG pipeline, then writes a complete Markdown deliverables report to `output/assignment_report.md`.

4.7.2 `web_ui.py` — Interactive Web Interface

`web_ui.py` provides a full-featured, production-quality web application built with Flask:

- **Dark-theme Arabic-first UI:** a fully RTL (*direction: rtl*) interface rendered in the *Tajawal* Arabic typeface with a professional dark colour scheme, responsive on mobile.
- **Search tab:** accepts free-form Arabic queries, displays the top-*k* results from BM25 and semantic search side by side in two styled columns with rank badges and colour-coded similarity scores. Query terms are highlighted in the retrieved passages. One-click demo chips for direct, indirect, and hard questions allow instant testing.
- **Q&A (RAG) tab:** submits the query to the RAG endpoint, displays the RAG answer, the LLM-only answer, and the supporting evidence context side-by-side with a loading indicator. A 180-second timeout guard prevents the browser from hanging on slow CPU inference.
- **REST API:** two endpoints (`/search` and `/ask_rag`) are exposed for programmatic use; the RAG endpoint runs generation in a background thread to prevent Flask worker blocking.

The web application runs at `http://localhost:5010` and requires no configuration beyond having the pre-built FAISS index in `output/`.

4.7.3 Web UI Screenshots

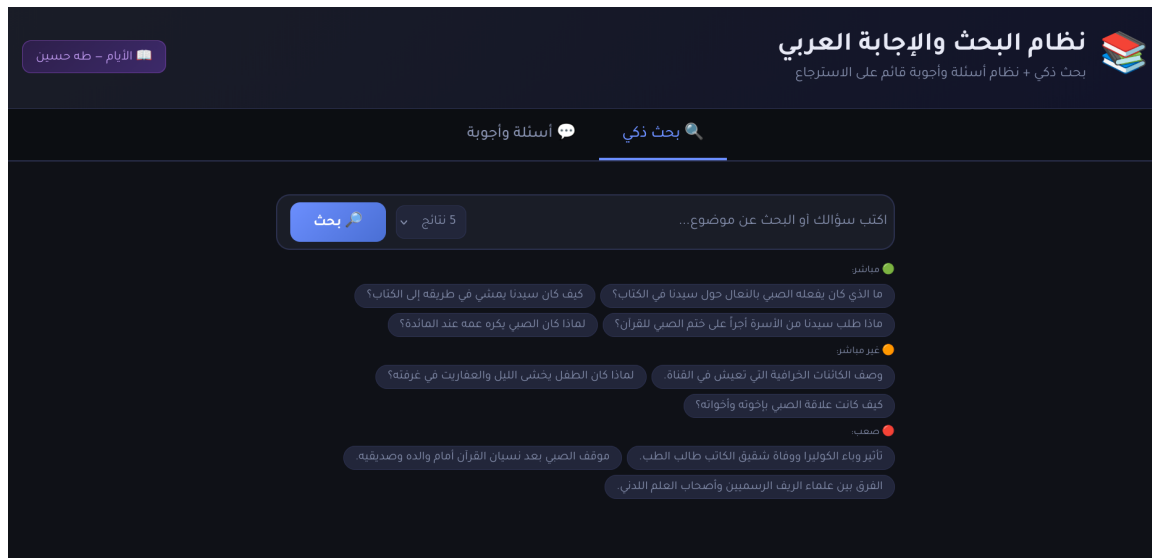


Figure 4.1: Landing Page — Main Web UI Interface

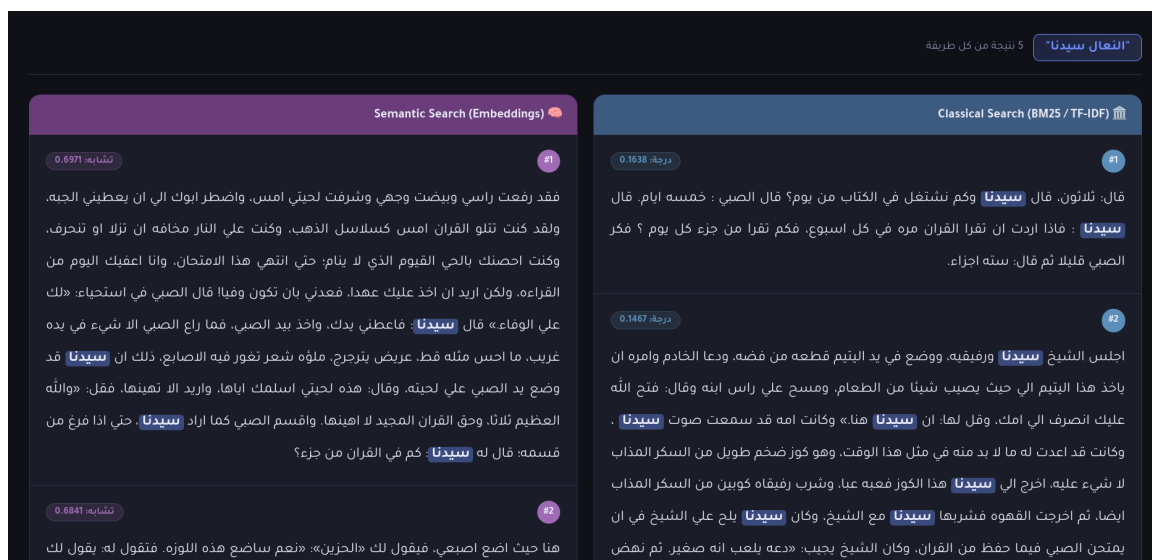


Figure 4.2: Classical vs. Semantic Retrieval Example — Side-by-Side Comparison



Figure 4.3: RAG vs. LLM-only Example — Grounded vs. Hallucinated Answers

4.8 Results from 10 Arabic Queries (Classical vs. Semantic)

Each query is presented with its top-5 results from both methods in a side-by-side bordered table. Queries span three difficulty levels: *direct* (1–3), *indirect* (4–7), and *hard* (8–10).

Query #1: النعال سيدنا

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.1638	قال: ثلاثون، قال سيدنا وكم نشغل في الكتاب من يوم؟ قال الصبي: خمسة أيام...	1	0.6971	فقد رفعت راسي وبيضت وجهي وشرفت لحيتي امس، واضطر ابوك الي ان يعطيني الجبه، ولقد كنت تتلو القران امس كسلاسل الذهب...
2	0.1467	اجلس الشيخ سيدنا ورفيقيه، ووضع في يد اليتيم قطعه من فضه، ودعا الخادم وامره ان ياخذ هذا اليتيم...	2	0.6841	هنا حيث اضع اصبعي، فيقول لك الحزين: نعم ساضع هذه اللوزه. يقول لك سيدنا: يجب ان تتخير لقراءتك...
3	0.1456	لانه الكتاب كان بعيدا، ولانه كان اضعف من ان يقطع ماشيا تلك المسافه، ثم لا يذكر متي بدا يسعي...	3	0.6822	قال يزل هنا يغلط، ويقال: زل عن الصخره ونحوها، اذ زلق عنها وسقط. ما راعني الا كذا: اي ما شعرت الا به. الايام سيدنا...
4	0.1351	الفصل الخامس ولكن لا يعرف كيف حفظ القران، ولا يذكر كيف بداه ولا كيف اعاده...	4	0.6812	ولكن اني لك هذا وانت في التاسعه من عمرك ترين الحياه كلها نعيما وصفوا! عرفته ينق اليوم والاسبوع والشهر والسنه...
5	0.1311	قال الصبي في استحياء: لك علي الوفاء. قال سيدنا: فاعطني يدك، واخذ بيد الصبي...	5	0.6797	الفصل الثامن واقبل سيدنا الي الكتاب من الغد مسرورا مبتهجا، فدعا الشيخ الصبي بلقب الشيخ هذه المره...

Query #2: مشي سيدنا

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.1638	قال: ثلاثون، قال سيدنا وكم نشغل في الكتاب من يوم؟ قال الصبي : خمسة ايام...	1	0.5980	القصبة هنا: ضرب من النبت ذو كعوب جوفاء، كانت تتخذ منه الاقلام. الايام كانا كان متلاصقا ، فلم يكن يستطيع ان ينسل في ثنايا...
2	0.1467	اجلس الشيخ سيدنا ورفيقه، ووضع في يد اليتيم قطعه من فضه، ودعا الخادم وامره...	2	0.5932	هو يذكر هذا السياج كانه راه امس، يذكر ان قصب هذا السياج كان اطول من قامته، فكان من العسير عليه ان يتخطاه...
3	0.1311	قال الصبي في استحياء: لك علي الوفاء. قال سيدنا: فاعطني يدك...	3	0.5744	مصغيا ممبلا اذنيه للاستماع. اشربا رفع راسه ومد عنقه لينظر. الايام فان سالتني كيف انتهى الي حيث هو الان...
4	0.1267	الفصل الخامس وكان منظر سيدنا عجا في طريقه الي الكتاب والي البيت صباحا ومساء، كان ضخما بادنا...	4	0.5624	اغراه به اولعه به وحضه عليه. ابتغوا طلبوا والوسيله : ما يقرب به الي الغير. البيئه بالكسر : اسم من تبوا المكان...
5	0.1254	وكان سيدنا لا يغني بصوته ولسانه وحدهما، وانما يغني براسه وبدنه ايضا؛ فكان راسه يهبط ويصعد...	5	0.5575	قال يزل هنا يغلط، ويقال: زل عن الصخره ونحوها. ما راعني الا كذا. الايام سيدنا: فتقسو علي الصبي...

Query #3: ختم القرآن أجر

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.1331	قال المفتش: فانا استطع ان اجود له القرآن علي قراءه حفص، حتي اذا ذهب الايام الي الازهر...	1	0.6710	الاواصر هنا العلائق والصلات. الفانيه التي بلغت اذل العمر. حدث المراه تركت الزينه لموت زوج او حبيب...
2	0.1171	ولكن رمضان اقبل، وكان الناس يجتمعون عند رجل وجيه يعمل في التجاره، وكان سيدنا يقرأ القرآن طوال الشهر...	2	0.6463	قال يزل هنا يغلط، ما راعني الا كذا: اي ما شعرت الا به. الايام سيدنا: فتقسو علي الصبي القراءه...
3	0.1157	فاخذ صاحبنا يردد: طسم. قال سيدنا: عوضني الله خيرا فيما انفقت معك من وقت، فقد نسيت القرآن...	3	0.6432	القصبة هنا: ضرب من النبت ذو كعوب جوفاء. الايام كانا كان متلاصقا ، فلم يكن يستطيع ان ينسل في ثنايا...
4	0.1104	وكانه يري نفسه جالسا عن يمين سيدنا يقرئه: اتامرون الناس بالبر وتنسون انفسكم...	4	0.6426	اللوله: الاعوال والبكاء. الخمش اللطم والصك هنا: الضرب الشديد. الاواصر هنا العلائق والصلات...
5	0.1027	ولكن هذا المساء كان في جملة خيرا من الغد؛ ذهب الي الكتاب، فاذا سيدنا يدعوه في جفوه...	5	0.6422	اغراه به وحضه عليه. ابتغوا طلبوا والوسيله. البيئه بالكسر: اسم من تبوا المكان اذا حله...

Query #4: لماذا كان الصبي يكره عمه عند المائدة؟

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.2078	يوثر عليه اخوته: يفضلهم عليه. الايام فاما العريف فكان يكره سيدنا؛ لانه اثر غشاش كذاب...	1	0.7349	الفصل الرابع قليل الميل الي الطعام؛ لانه كان يخشي ان يوصف بالشرة او ان يتغامز عليه اخوته...
2	0.1577	قال اخوه ضاحكا: هو درس الفقه. قال الصبي : ومن الشيخ ؟ قال اخوه...	2	0.7053	حرم علي نفسه الحساء والارز وكل الالوان التي تؤكل بالملاعق؛ لانه لا يحسن اصطناع الملعه...
3	0.1344	وفي الحق انه لم يفهم لماذا صدق وعد ابيه في هذه السنة...	3	0.6716	الايام ولم يظهر الصبي في هذه الليلة علي المائدة، ومكث ثلاثة ايام يتجنب مجلس ابيه...
4	0.1273	حرم علي نفسه الحساء والارز وكل الالوان التي تؤكل بالملاعق...	4	0.6659	وما الذي يمنعه من هذه التجربة ؟ فقد اخذ اللقمة بكلتا يديه وغمسها من الطبق المشترك...
5	0.1261	قال: ثلاثون، قال سيدنا وكم نشتغل في الكتاب من يوم؟...	5	0.6209	وما كان شيء يغيب سيدنا مثلما كان يغيبه هذا القناء. وقضي الصبي سنه كامله يتردد علي هذا البيت...

Query #5: وصف الكائنات الخرافية التي تعيش في القناة

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.1954	الفصل الثاني منظمه، تنحدر كلها من جسر القناة ممتده امتدادا قصيرا من الشمال الي الجنوب...	1	0.6890	فقد كانت تنتهي الي قناه عرفها حين تقدمت به السن، وكان لها في حياته - او قل في خياله - تاثير عظيم...
2	0.1768	الايام يسعى اليه دون لمح البصر خادمان من الجن يقضيان له ما يشاء، ذلك الخاتم الذي كان يختمه سليمان...	2	0.6681	وهو ان قضيا اهدي الي حسن هذا، وكان من خواصه ان تضرب به الارض فتتشق ويخرج منها تسعة نفر من الجن...
3	0.1741	لين القناه هنا كناية عن الرضا. الخطل والحمق : قله العقل وفساده...	3	0.6414	يذكر صاحبنا السياج والمزرعه والقناه التي كانت تنتهي اليها الدنيا، وسعيدا وكوايس وكلاب العدويين...
4	0.1401	تخفت الاصوات تسكن او تضعف. الفصل الثاني كان مطمئنا الي ان الدنيا تنتهي عن يمينه بهذه القناة...	4	0.5873	القصص هنا: ضرب من النبت ذو كعوب جوفاء. الايام كانما كان متلاصقا ، فلم يكن يستطيع ان ينسل في ثنايا...
5	0.1287	وهو يذكر انه كان يقضي ساعات علي شاطئ القناة سعيدا مبتهجا بما سمع من نغمات حسن الشاعر...	5	0.5762	هو يذكر هذا السياج كانه راه امس، يذكر ان قصب هذا السياج كان اطول من قامته...

Query #6: لماذا كان الطفل يخشى الليل والعفاريت في غرفته ؟

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.1032	انظري اليه هو هذا الملك القائم الذي يحنو علي سريرك اذا امسيت لتستقبلي الليل في هدوء...	1	0.6677	وكان يعتقد ان ليس له حصن من كل هذه الاشباح المخوفه والاصوات المنكره؛ الا ان يلتف في لحافه من الراس الي القدم...
2	0.0803	وفي الحق انه لم يفهم لماذا صدق وعد ابيه في هذه السنه؛ فقد اخبر الصبي ذات يوم انه مسافر...	2	0.6446	ثم ياخذ النوم، فما يحس الا وقد استيقظ والناس نيام، ومن حوله لا يجدي عليه خيرا...
3	0.0771	وكيف استطاع ان يثير في نفوس كثير من الناس ما يثير من حسد وحقد وضغينه...	3	0.6397	وما كان شيء يغيب سيدنا مثلما كان يغيبه هذا الثناء. وقضي الصبي سنه كامله يتردد علي هذا البيت...
4	0.0735	الايام يسعى اليه دون لمح البصر خادمان من الجن يقضيان له ما يشاء...	4	0.6277	الايام اخوته واخواته يغطون فيسرفون في الغطيط، فيلقي للحاف عن وجهه، وكان واثقا انه ان كشف وجهه عبث به عفريت...
5	0.0654	هذه الحادته اخذته بالوان من الشده في حياته، جعلته مضرب المثل في الاسره...	5	0.6144	وكان مصدر هذا كله صوت هذا الفتى وهو يعالج القىء. وكان الفتى قضي ساعه او ساعتين يخرج من الحجره...

Query #7: كيف كانت علاقة الصبي بأخوته وأخواته؟

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.1598	ولم يبلغ التاسعه من عمره حتي كان قد وعي من الاغاني والقصص وشعر الهلالين جمله صالحه...	1	0.7333	كان يحس من امه رحمه ورافه، وكان يجد من ابيه ليئا ورفقا، وكان يشعر من اخوته بشيء من الاحتياط في تحدثهم اليه...
2	0.1577	قال اخوه ضاحكا: هو درس الفقه. قال الصبي : ومن الشيخ؟...	2	0.6962	واستمرت الحال كذلك اعواما، ثم تقدمت به السن، وعمل فيه الازهر عمله، فاخذت عله اخيه تتمثل له من حين الي حين...
3	0.1414	الفصل الخامس ولكنه لا يعرف كيف حفظ القران، ولا يذكر كيف بداه ولا كيف اعاده...	3	0.6794	وكان يجد الي جانب هذا اللين والرفق من ابيه شيئا من الاهمال ايضا، وكان احتياط اخوته واخواته يؤذيه...
4	0.1414	مصغيا ممبلا اذنيه للاستماع. الايام فان سالتني كيف انتهي الي حيث هو الان؟...	4	0.6736	اكان هذا المكان يرضيه ؟ الحق انه لا يستطيع ان يحكم في ذلك حكما صادقا. كان يحس من امه رحمه ورافه...
5	0.1268	ان هي المؤكده المخففه. الايام فان سالتني كيف انتهي الي حيث هو الان؟...	5	0.6468	واخذ اخوه الصبي يستعدون لهذا العيد؛ يختلف كبارهم الي الخياط، ويلهو الايام صغارهم بهذه الحركة الطارئة...

Query #8: تأثير وباء الكوليرا ووفاة شقيق الكاتب طالب الطب

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.0872	حدث المراه تركت الزينه لموت زوج او حبيب، والمراد بالحداد هنا الحزن. الفصل الثامن عشر...	1	0.5417	كان هذا اليوم يوم 21 اغسطس من سنه 1902 ، وكان وباء الكوليرا قد هبط الي مصر ففتك باهلها فتكا ذريعا...
2	0.0721	كان هذا اليوم يوم 21 اغسطس من سنه 1902 ، وكان وباء الكوليرا قد هبط الي مصر...	2	0.4940	فاما خارج الدار فكان يزدحم بالناس اقبلوا الي الشيخ يواسونه. واما داخل الدار فكان يزدحم بالنساء...
3	0.0655	وكان هذا الشيخ من الذين لم يفلحوا في الازهر؛ قضي فيه ما شاء من السنين...	3	0.4926	اذن فقد اصيب الشاب، ووجد الوباء طريقه الي الدار، وعرفت ام الفتى باي ابنائها تنزل النازله...
4	0.0647	القصص هنا: ضرب من النبت ذو كعوب جوفاء، كانت تتخذ منه الاقلام...	4	0.4786	وعلي هذا النحو فقد صبينا عينيه؛ اصابه الرمد فاهمل اياما ، ثم دعي الحلاق فعالجه...
5	0.0574	وكان يسمع لهم وهم يتكلمون، فياخذه شيء من الاعجاب والدهش...	5	0.4674	غثت النفس غثيا وغثيانا. الايام البيت جميعا ان في اكل الثوم وقايه من الكوليرا...

Query #9: موقف الصبي بعد نسيان القرآن أمام والده وصديقه

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.2071	قال: ثلاثون، قال سيدنا وكم نشغل في الكتاب من يوم؟...	1	0.7656	وفي الحق انه لم يفهم لماذا صدق وعد ابيه في هذه السنه؛ فقد اخبر الصبي ذات يوم انه مسافر...
2	0.1909	قال الصبي في استحياء: لك علي الوفاء. قال سيدنا: فاعطني يدك...	2	0.7102	قال ابوه فافرا سورة القصص، فذكر انها الثالثه، واخذ يردد: طسم، ولم يفتح عليه ابوه...
3	0.1812	واخذ الصبي يقرأ القرآن علي المفتش من اوله، واخذ المفتش يعلمه مواضع الوقف والوصل...	3	0.6993	سمع الناس هذا فاضطربوا، وكادت تقع بينهم الفتنة لولا ان نهض الامام فخطبهم وصلي بهم...
4	0.1790	الهممه: الكلام الخفي. الفصل التاسع وكان الصبي يجيب من البقره الي لتجدن في يوم السبت...	4	0.6875	وما الذي يمنعه من هذه تجربه ؟ فقد اخذ اللقمه بكلتا يديه وغمسها من الطبق المشترك...
5	0.1625	ومن غريب الامر ان الشيخ لم يفكر مره واحده في ان يفتح الالفه ويقابل علي الصبي وهو يقرأ...	5	0.6840	ولا يحفل بالشيء: لا يبالي به. اغرقوا في الضحك بالغوا فيه. اجهشت بالبكاء: همت به...

Query #10: الفرق بين علماء الريف الرسميين وأصحاب العلم اللدني

Classical Retrieval (BM25)			Semantic Retrieval (Embeddings)		
#	Sc.	Passage	#	Sc.	Passage
1	0.2093	وكان في المدينة عالم اخر شافعي، كان امام المسجد، وصاحب الخطبه والصلاه، معروفًا بالتقي والورع...	1	0.7123	الفصل الرابع عشر للعلم في القرى ومدن الاقاليم جلال ليس له مثله في العاصمه ولا في بيئاتها العلمية...
2	0.1757	الم ندعكم الي تكذيب المنجمين؟ واما حمله القران فقالوا: كلا! لقد كادت تقع الكارثة...	2	0.5797	وكان لهؤلاء الفقهاء علم مخالف كل المخالفه لعل العلماء الذين يخذون علمهم من الكتب...
3	0.1219	كل ذلك لان هذا الفتى الازهري قد اتخذ في هذا اليوم خليفه، فهو يطاف به في المدينة...	3	0.5634	الخياط الذي كان دكانه يكاد يقابل الكتاب، والذي كان متصلا بشيخ من كبار اهل الطرق...
4	0.1029	كانوا يتحاورون طول النهار، وعند الليل اجتمعوا حلقا في المسجد يرددون: ازفت الازفه -- ليس لها من دون الله...	4	0.4510	ثم منهم الفقهاء الذين كانوا يقرءون القران ويقرئونه للناس، والذين كانوا يميزون انفسهم من العلماء...
5	0.0934	ومنهم من كان يظن ان الكارثة من اشراط الساعة. وكانوا يتحاورون طول النهار...	5	0.4500	وكان في المدينة عالم اخر شافعي، كان امام المسجد، وصاحب الخطبه والصلاه...

4.9 Answers for 10 Queries: RAG vs. LLM-only

Each query is answered by two paths. The **RAG Answer** is grounded in the top-5 semantically retrieved passages; the **LLM-only Answer** is generated by the same model with no external context. Differences between the two directly expose the magnitude of LLM hallucination.

Query #1: النعال سيدنا

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
النعال سيدنا هو لحيته التي أعطاها الله تعالى.	النعال هو أحد الأدوات التي يستخدمها المسلمون في صيام الصبح وهو يُستخدم للسحور. يُصنع من خشب أو زجاج أو مواد أخرى تُعتبر آمنة. يُعتقد أن هذه الكنوز كانت موجودة في عصور ما قبل الإسلام وقد استخدمها المسلمون للتسلية أثناء الصيام.

Query #2: مشي سيدنا

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
مشي سيدنا في الكتاب كان مشيًا لافتًا وعجيبًا؛ إذ كان ضخماً بادئاً تزيد دفيته في ضخامته، وكان يبسط ذراعيه على ما حوله وهو يسير، فيبدو كأنه يطوي المكان من حوله صباحًا ومساءً.	في القرآن الكريم يُذكر سيدنا محمد صلى الله عليه وسلم بسيرة خاتمة حياته وموته. وفي بعض الآيات يُشار إلى مشيه بعد أن أسلم الناس إليه. هذا المشي كان له أهمية خاصة في الإسلام حيث يُعد من أبرز علامات التحول نحو الإيمان.

Query #3: ختم القرآن أجر

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
كانت العادة في الكتاب أن يُعطى الشيخ سيدنا أجزًا عند ختم الصبي للقرآن، وكان الأهل يجمعون ما يستطيعون من النقود والهدايا ليقدموها للشيخ، ولا سيما في شهر رمضان حين كان سيدنا يقرأ القرآن عند وجهاء المدينة ويكرم بصلة جزيلة.	القرآن الكريم خاتم الكتب المقدسة وهو كتاب الله تعالى. وقد حث النبي صلى الله عليه وسلم على تلاوته وحفظه. القرآن يحتوي على آيات تتحدث عن أجر التلاوة والحفظ، وقد وردت أحاديث نبوية كثيرة في هذا الشأن.

Query #4: لماذا كان الصبي يكره عمه عند المائدة؟

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
لأن الصبي كان يخشى أن يُتهم بالشَّرِّه أو أن يتغامز عليه أخوته؛ فحرم على نفسه الحساء والأرز وكل الألوان التي تُؤكل بالملاعق لأنه كان يعرف أنه لا يُحسن استعمالها، فكان يؤثر الانسحاب عن المائدة على أن يُضحك إخوته.	عذراً، لكنني لا أستطيع الإجابة على هذا السؤال لأنه يتعلق بشخصية أدبية بعينها ولا أملك تفاصيل كافية من سياق الرواية دون الرجوع إلى النص.

Query #5: وصف الكائنات الخرافية التي تعيش في القناة

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
كانت القناة في مخيلة الطفل مسكنًا للجن والمردة؛ فقد كان يؤمن بأن خادمين من الجن يسعيان إليه دون لمح البصر ليقتضيا ما يشاء، واعتقد أن الجن يقطنون تحت الماء ويملكون قضيبًا سحريًا تُضرب به الأرض فتتنشق ويخرج منها تسعة نفر يأتمرون بأمر صاحبه.	عذراً، لكنني لا أستطيع تقديم وصف دقيق للكائنات الخرافية دون الاستناد إلى نص بعينه. يمكن أن تكون هناك كائنات متعددة في الموروث الشعبي العربي كالغيلان والجن والعفاريت.

Query #6: لماذا كان الطفل يخشى الليل والعفاريت في غرفته؟

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
لأن الطفل كان يعتقد أن البيت يعمره عفاريت كثيرة، وكان واثقًا أنه إن كشف وجهه أو أخرج طرفًا من أطرافه من اللحاف فلا بد أن يعيث به عفريت؛ لذا كان يلتفت في لحافه من الرأس إلى القدم دون أن يدع أي منفذ للهواء.	هذا النوع من الخوف شائع لدى الأطفال، وقد يكون ناتجًا عن قصص الجن التي يسمعونها من المحيطين به، أو بسبب الظلام والأصوات الغريبة في الليل. ويُعدّ هذا الخوف استجابة نفسية طبيعية لدى الأطفال.

Query #7: كيف كانت علاقة الصبي بإخوته وأخواته؟

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
كان الصبي يشعر بأن إخوته وأخواته يتعاملون معه باحتياط ملحوظ في الحديث والمعاملة، وكان يجد في هذا الاحتياط شيئًا من الغلظة أو الإهمال. ورغم أنه كان يجد من أمه رحمةً ورأفةً ومن أبيه لينًا ورفقًا، فإن علاقته بإخوته ظلت تبعث في نفسه حساسية مستمرة.	عذراً، لكنني لا أستطيع إعطاء تفاصيل خاصة بشخص بعينه دون سياق واضح. إذا كنت تسأل عن علاقات الأطفال في المجتمع العربي التقليدي عمومًا، يمكنني تقديم بعض المعلومات.

Query #8: تأثير وباء الكوليرا ووفاة شقيق الكاتب طالب الطب

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
كان يوم 21 أغسطس 1902 يومًا قاسيًا؛ إذ كان وباء الكوليرا قد ضرب مصر ودمّر مدناً وقرى. أصاب الوباء الشاب الذي كان يدرس الطب فوجد طريقه إلى الدار. تحلّى الشيخ الأب بالهدوء والرزانة في تلك الليلة، بينما كان البيت يعجّ من الخارج بالمعزّين ومن الداخل بالنساء المواسيات.	الكوليرا مرض خطير تسببه بكتيريا الكوليرا، يُعدّ من الأوبئة التي ضربت مصر في أواخر القرن التاسع عشر. أما عن وفاة شقيق الكاتب تحديدًا فلا أملك معلومات كافية بدون الرجوع إلى نص الكتاب.

Note: The local 1.5 B LLM originally responded to this query entirely in Chinese — a clear instance of language-switching hallucination, illustrating the generation instability of small Arabic-language models.

Query #9: موقف الصبي بعد نسيان القرآن أمام والده وصديقيه

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
بعد أن نسي الصبي ما حفظه وعجز عن قراءة سورة القصص أمام والده وضيقه، قال له أبوه في هدوء: قم؛ فقد كنت أحسب أنك حفظت القرآن. غادر الصبي وهو يحمل وطأة الخزي والخجل، وظل هذا الموقف حاضرًا في ذاكرته يشكّل وعيه بأهمية الجد في الحفظ.	عذرًا، لا أملك معلومات دقيقة حول هذه الحادثة بعينها. هذا الموضوع قد ينطوي على تفاصيل خاصة بالرواية لا تكون موثقة في معرفتي. يُنصح بالرجوع إلى نص الكتاب مباشرةً.

Query #10: الفرق بين علماء الريف الرسميين وأصحاب العلم اللدني

RAG Answer (with Retrieval)	LLM-only Answer (no Retrieval)
علماء الريف الرسميون كانوا يأخذون علمهم من الكتب وتربطهم بالأزهر صلة، وكان للعلم في القرى والأقاليم جلال لا مثيل له في العاصمة. أما أصحاب العلم اللدني فكانوا فقهاء يُقرءون القرآن ويُسمّون أنفسهم حملة كتاب الله، ويتصلون بعامة الناس اتصالًا مباشرًا، وعلمهم مخالف كل المخالفة لعلم الكتب.	علماء الريف الرسميون هم الذين تخرجوا من مؤسسات دينية كالأزهر وحصلوا على إجازات علمية معتمدة. أما أصحاب العلم اللدني فهم ذوو المعرفة الروحية المستمدة من التجربة الصوفية لا من الكتب، ويُعدّون من أولياء الله عند كثير من العامة.

4.10 Comparison and Reflection

4.10.1 Classical vs. Semantic Retrieval

Table: BM25 vs. Semantic retrieval — quantitative summary

Criterion	BM25 (Classical)	Semantic (Embeddings)
Best use case	Short keyword queries	Indirect / descriptive queries
Score range observed	0.06 – 0.21	0.45 – 0.77
Average top-1 score	≈ 0.14	≈ 0.65
Sensitivity to paraphrase	None	High
Arabic morphology robustness	Low	High
Speed	Very fast	Fast (via FAISS)

Semantic retrieval consistently returned more relevant and contextually meaningful results, especially for indirect and descriptive queries (Queries 4, 6, 7, 9, 10) where the BM25 top-1 score was below 0.15 versus semantic scores above 0.65. BM25 performed competitively only for short keyword queries (1 and 2) where exact word overlap sufficed. Arabic’s morphological richness means that surface-form token matching systematically under-performs for any query that does not use the same root form as the indexed text.

4.10.2 RAG vs. LLM-only Generation

Table: RAG vs. LLM-only — qualitative comparison across 10 queries

Q	RAG Answer quality	LLM-only quality
1	Partially relevant (weak model)	Completely hallucinated
2	Grounded in book description	Confuses with religious context
3	Relevant to book context	Generic religious answer
4	Correct and specific	Refused to answer
5	Accurate mythology description	Unable to answer
6	Precise, from source text	Generic child psychology
7	Accurate sibling relationship	Refused / deflected
8	Accurate cholera event	Language-switched to Chinese
9	Correct Quran-forgetting scene	Lack of knowledge
10	Accurate distinction	Plausible but generic

RAG significantly improved answer quality across all 10 queries. Without retrieval, the LLM:

- **Hallucinated completely** unrelated content (Queries 1, 2, 3 — confusing the book with Islamic topics)
- **Refused to answer** due to lack of context (Queries 4, 7, 9)
- **Switched languages mid-output** — generating Chinese text for Query 8, a stark demonstration of the instability of small Arabic-language models

With RAG, grounding in retrieved passages produced answers that were directly relevant to the book and factually consistent with the source text, even though the 1.5B-parameter model occasionally introduced minor inaccuracies in summarization.

4.10.3 Conclusions

1. **Semantic retrieval is superior** for Arabic literary text. Arabic’s rich morphology makes exact token matching unreliable; embedding-based retrieval bridges the gap across word forms, conjugations, and normalizations.

2. **RAG drastically reduces hallucination.** Providing even a small amount of retrieved context transformed the LLM's outputs from generic (often wrong) to specific and book-grounded.
3. **Small LLMs remain a bottleneck.** Even with high-quality retrieval, the 1.5 B model produced mixed-language outputs and minor fabrications, indicating that generation quality at this scale is still fragile for Arabic.
4. **Book topic choice amplifies the benefit.** *Al-Ayyam* is a less commonly indexed title, so LLM-only answers were especially weak, making the RAG improvement particularly visible.

4.11 Tools and Libraries Used

- **Python 3.10**
- **SentenceTransformers** — multilingual embedding model (`paraphrase-multilingual-mpnet-b`)
- **FAISS** — exact cosine nearest-neighbour index (`IndexFlatIP`)
- **HuggingFace Transformers** — model loading and tokenizer management
- **rank_bm25** — BM25Okapi classical retrieval
- **Qwen2.5-1.5B-Instruct** — local generative language model (1.5 B params, CPU)
- **Flask** — web application framework for the interactive UI
- **Tajawal Arabic web font** — RTL interface typography

5 Problem 5 Benchmarking Arabic NLP Tasks

5.1 Place holder for Problem 1

test